Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · LLM Applications · RAG Workflows · AI Agents · Evals & Guardrails · Responsible AI · Python · Production

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AgentProbe (live eval toolkit) | Fairfax, Virginia · Open to Onsite NYC

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production experience, specializing in **forward deployed AI engineering, LLM applications, RAG workflows, and agentic systems with evals and guardrails built in from day one**. Currently at Intuit owning end-to-end: discovering AI use cases, building LLM applications with **prompt management, tool use, and RAG grounding**, then productionizing them with guardrails, LLM-as-judge evals, monitoring, logging, and fallback paths -- zero hallucination incidents post-deploy, eval harness catches regressions in CI before release. Built **Leasel**: an 8-agent LangGraph platform with **confidence-gated HITL escalation, responsible AI enforced in code** (LLM reasons, human or deterministic logic decides every regulated call), and adversarially tested safety boundaries -- 93.6% precision, 87.9% F1 across 1,240 test cases. Built **AgentProbe**: open-source LLM evaluation toolkit with a 5-metric eval engine, RAG quality scoring, and CI regression detection. 7+ years at Entrata (100K+ MAU) shipping production systems across 120+ Agile sprints -- cross-functional delivery with Legal, Compliance, and Operations at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Forward Deployed AI Engineering | LLM Applications & Automation Tools | AI Agents & Agentic Systems | RAG Workflows & Retrieval | Evals & LLM-as-Judge | Guardrails & Responsible AI | HITL & Human Oversight | Prompt Management & Tool Use | Model Governance & Privacy | Monitoring & Logging | Python & FastAPI | AWS & Production AI at Scale

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — LLM Applications · Guardrails · Evals · Python · AWS

- Discover, prototype, and **productionize LLM applications** end-to-end: financial summaries, natural-language query answering, and reporting features -- own each use case from discovery through launch, monitoring, and iteration; ground Claude responses in customer data via **RAG workflows** and retrieval for full citation traceability.
- Build **production guardrails** around LLM calls: input validation, sensitive-data redaction, output schema enforcement, and **fallback paths** when model output fails checks -- responsible AI enforced at the code layer; implement **prompt management** and structured-output schemas so financial answers stay consistent and safe.
- Build **LLM evaluation harness** with LLM-as-judge scoring, golden datasets, and CI regression detection; caught 3 prompt regressions before release, zero hallucination incidents post-deployment; establish **model governance** and privacy standards across all Claude-powered APIs; add **monitoring and logging** via OpenTelemetry for production AI visibility.
- Build ML pipelines (scikit-learn, XGBoost) for financial predictions; develop React/TypeScript front ends; containerize services with Docker and ship through GitHub Actions to AWS with full observability.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · AWS · PostgreSQL · Distributed Systems

- Shipped 20+ production features end-to-end (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints; cross-functional delivery with Legal, Compliance, Operations, and Product; reviewed 300+ PRs; mentored junior engineers.
- Built **automation tools** in Python/Bash eliminating 80% of manual operational work; built internal debugging and data investigation tools cutting triage time 40%; reduced 200+ SQL queries' runtime by 30% via execution plan analysis and indexing.
- Owned HUD income-limit ETL pipeline for 7 years: few thousand federal rows → 180K domain records with validation, rollback checks, and compliance enforcement; zero violations across 7 annual release cycles -- income limit errors affect resident housing eligibility, so correctness is non-negotiable.
- AWS S3 PHI migration with **Libsodium encryption** (100% PHI compliance, zero security incidents); Docker containerization cutting microservices execution time 50%; RabbitMQ async processing with idempotency guarantees for bulk tenant record updates.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · PyTorch · HuggingFace · Statistical Modeling

- Built predictive models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection; built fine-tuned classification model with HuggingFace + PyTorch (presented to faculty panel). Mentored 600+ students across Python and database design.

PORTFOLIO PROJECTS

Leasel — Forward Deployed AI Platform with HITL & Responsible AI

Forward Deployed AI · Agentic AI · Guardrails

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Owned end-to-end delivery across 3 product surfaces (Document Intelligence, Support Ticket Investigation, Tenant Self-Service) sharing one SQLite knowledge repo -- **responsible AI enforced in code**: LLM reasons; human or deterministic validator decides every regulated call; safety boundary adversarially tested -- never auto-resolves eligibility or financial decisions for a tenant.
- 8-agent LangGraph StateGraph with real conditional edges and **confidence-gated HITL escalation**: triage → retrieval → config/history/structured-data agents → reasoning → validation → human; correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release.
- **RAG workflow** (hand-rolled TF-IDF + cosine reranking + section-aware chunking): 100% citation correctness, 0% unsupported-claim rate; **evals** via AgentProbe LLM-as-judge CI harness: 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe — LLM Evaluation Toolkit & RAG Quality Monitor

Evals · LLM-as-Judge · Monitoring

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression detection blocks production deploys when quality drops below threshold; 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) handle multi-step evaluation and auto-remediation.
- 15-endpoint FastAPI backend routing **LLM applications** across 5 providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) via LLM Router with **tool use** and dynamic task classification; 100% local, zero vendor lock-in.

AI on FHIR — Clinical NLP with HIPAA Guardrails & Responsible AI

Responsible AI · HIPAA Compliance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP pipeline translating English/Spanish clinical questions into FHIR API calls using LLMs, BERT, and embeddings; **HIPAA guardrails**: role-based access control for data isolation, output filtering for PHI; ~60% reduction in manual clinical lookup time; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI & LLMs: LLM Applications, AI Agents, Agentic AI, Forward Deployed AI Engineering, LLM Guardrails, Responsible AI, Model Governance, Evals, LLM-as-Judge, Prompt Management, Tool Use, HITL, Human-in-the-Loop, RAG Workflows, Retrieval-Augmented Generation, Automation Tools, Hallucination Detection, Fallback Paths, LangGraph, LangChain, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Prompt Engineering, Structured Outputs

Retrieval & NLP: RAG Pipelines, Hybrid Search, Similarity Search, Vector Database, ChromaDB, Pinecone, Weaviate, Elasticsearch, TF-IDF, Embeddings, Semantic Search, BERT, HuggingFace, spaCy, Document Intelligence, OCR

Monitoring & LLMOps: Monitoring, Logging, OpenTelemetry, Prometheus, Grafana, LLMOps, CI/CD for LLM Systems, Regression Detection, Golden Datasets, Eval-Driven Development, GitHub Actions

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark, Java, Spring Boot, TypeScript, Node.js, React, PHP, SQL, PL/SQL, Bash

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization, Stored Procedures

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, RabbitMQ, Kafka, Celery